How to Write Reports

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Abstract

This short article explains how to write the Final Project Report that is more than 10 pages excluding figures, tables and references. Including all, it may be about 15 pages (based on this template). This LeTex template can be used to write the report. If you are not proficient in using LeTex, there is no need to use latex and MS-WORD or any other paper writing tool can be used. The abstract summarizes your research and its outcomes without technical details.

1 Introduction

The introduction of the paper expands the abstract and list main contributions of the paper with respect to existing literature. This is where you do literature review explaining what similar works have been done by others. With that list of literature, you contrast your main contribution of the research. It is expected you list reasonable number of previous papers. Try to add at least 10-20 references and explain what is new or different about your method with respect to existing methods in literature. You need detailed review discussing cons and pros of existing methods against what you are trying to do. Also you need to able to spell out at least 2-3 major contributions or key points of your research.

2 Materials and Methods

2.1 Data

The section explains in detail what the data set is. Enough detailed explanation about data and preprocessing on the data is a must. For instance, you need to explain about what scanner was used and what is the image resolution and image acquisition parameters are.

2.2 Methods

Then you describe in great detail what methods you used in analyzing the data. It is expected you will be using methods we studied in the class.

2.3 Validation

The validation is designed to establish not only overall performance but also robustness under different error regimes, with particular emphasis on false positives and false negatives. All procedures will be described in sufficient detail, including simulation generation, preprocessing, parameter choices, and evaluation metrics, to ensure full reproducibility by independent researchers. Validation with ground truth will be performed using simulated data where the underlying structure is explicitly known. This allows direct quantification of how accurately the proposed method recovers the true signal or structure of interest. To assess false positives, we will consider settings in which the target structure is absent and evaluate whether the method incorrectly detects spurious patterns or signals. To assess false negatives, we will construct scenarios where the target structure is present and evaluate the extent to which the method fails to detect or recover it. These two complementary settings provide a principled way to characterize sensitivity and specificity of the method.

Comparisons will be conducted against appropriate baseline methods that represent existing or standard approaches to the same problem. These baselines will be implemented under comparable conditions, including consistent data representations, preprocessing steps, and evaluation criteria. Performance will be evaluated using metrics aligned with the scientific objective, including reconstruction

error, statistical significance, and clustering accuracy. The comparison framework will be explicitly defined to ensure that any observed improvements can be attributed to the methodological contributions rather than differences in implementation. For real data, validation will focus on reproducibility, stability, and interpretability. This includes assessing consistency across repeated measurements, robustness to noise or subsampling, and agreement with known or expected patterns. Together, the simulation-based and real-data validation provide complementary evidence supporting the effectiveness of the proposed method.

A well-constructed example of the validation framework is provided in the link [1], which reproduces simulation studies from (Chung et al., 2023, 2024) and demonstrates systematic comparisons across multiple baseline methods under both false positive and false negative settings.

What is considered as baseline? Baseline methods are any established, widely accepted, or conceptually natural approach that addresses the same task under comparable assumptions, and serves as a reference point for evaluating improvement in accuracy, stability, interpretability, or computational efficiency. In rigorous methodological work, baselines should *not* be arbitrary alternatives but must represent meaningful widely accepted competing paradigms that reflect how a practitioner would reasonably solve the problem without the proposed method. The role of the baseline is therefore not merely comparative but interpretive: it clarifies what specific advantage is gained by introducing the proposed new method.

In the context of spectral expansion using the Hodge Laplacian (Anand and Chung, 2023), several natural baselines arise depending on the level of structure being exploited. At the most basic level, one may consider generic orthonormal bases constructed through *Gram–Schmidt orthogonalization*. While mathematically valid, such bases are agnostic to the underlying topology and do not encode geometric or combinatorial structure. As a result, they provide a useful null comparison demonstrating what is lost when no structural prior is imposed. A more meaningful and commonly accepted baseline is *principal component analysis (PCA)*, which constructs an orthonormal basis adapted to variance maximization. PCA captures dominant modes of variation in the data but ignores higher-order interactions encoded in simplicial complexes, such as cyclic flows or higher-dimensional cavities. Comparing against PCA therefore isolates the contribution

[1] `https://github.com/laplcebeltrami/BMI768/blob/main/clustering/SIMULATION_clustering.mlx`

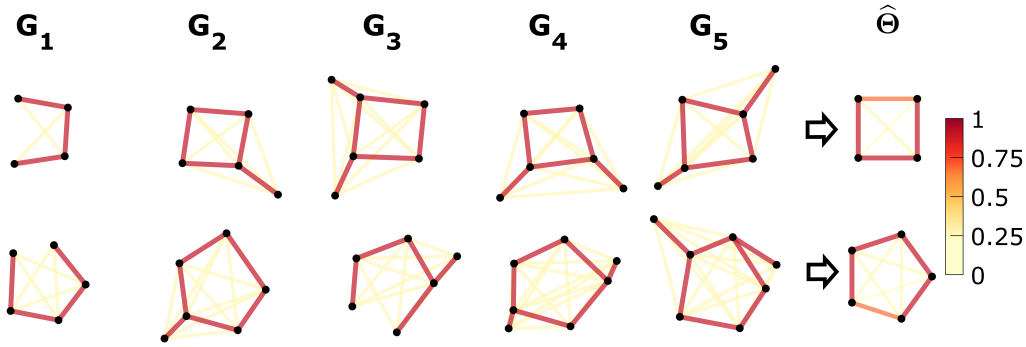

Figure 1: Figure illustrating how to average graphs of different sizes and topology using persistent homology. Generated by a former student who took the class (Songdechakraiwut and Chung, 2023).

of topology beyond second-order covariance structure.

Stronger baselines should include graph Laplacian based spectral methods, such as eigenvectors of the graph Laplacian (Chung, 2019). These methods incorporate pairwise relationships and are widely used in medical imaging and network analysis. However, they fundamentally operate on node signals (0-forms) and cannot distinguish between gradient and curl components of edge flows. In contrast, the Hodge Laplacian explicitly decomposes edge flows into gradient, curl, and harmonic components, enabling separation of local fluctuations from global topological structure. Thus, comparison with graph Laplacian methods directly demonstrates the benefit of modeling higher-order interactions and enforcing topological consistency.

3 Results

Results are often summarized using tables (Table 1), figures (Figure 1), plots and quantitative numbers. Simply displaying pretty figures is not results. Must have quantitative numbers to demonstrate your method. Also you need to provide an interpretation of the results. Must also provide interpretation of the results.

Table 1: Sample table summarizing the performance of the proposed topological loss $\mathcal{L}_{top}$ over other baseline methods. Generated by a former student who took the class (Songdechakraiwut and Chung, 2023).

d	c	p	GA	SM	RRWM	IPFP	$\mathcal{L}_{top}$
12 vs. 12	2 vs. 2	0.6	0.49 ± 0.27	0.46 ± 0.30	0.51 ± 0.30	0.47 ± 0.28	0.53 ± 0.29
		0.8	0.45 ± 0.25	0.47 ± 0.31	0.56 ± 0.29	0.47 ± 0.30	0.50 ± 0.30
	3 vs. 3	0.6	0.45 ± 0.32	0.44 ± 0.26	0.47 ± 0.27	0.51 ± 0.30	0.46 ± 0.31
		0.8	0.54 ± 0.31	0.51 ± 0.27	0.51 ± 0.29	0.52 ± 0.29	0.51 ± 0.30
	6 vs. 6	0.6	0.57 ± 0.30	0.51 ± 0.28	0.56 ± 0.29	0.45 ± 0.26	0.58 ± 0.29
		0.8	0.55 ± 0.29	0.48 ± 0.26	0.52 ± 0.27	0.54 ± 0.30	0.49 ± 0.27
18 vs. 18	2 vs. 2	0.6	0.48 ± 0.26	0.49 ± 0.32	0.54 ± 0.29	0.47 ± 0.30	0.54 ± 0.31
		0.8	0.52 ± 0.28	0.50 ± 0.28	0.46 ± 0.30	0.52 ± 0.25	0.50 ± 0.26
	3 vs. 3	0.6	0.49 ± 0.28	0.58 ± 0.31	0.43 ± 0.28	0.51 ± 0.27	0.53 ± 0.30
		0.8	0.46 ± 0.30	0.51 ± 0.27	0.52 ± 0.33	0.45 ± 0.29	0.53 ± 0.27
	6 vs. 6	0.6	0.53 ± 0.28	0.48 ± 0.30	0.51 ± 0.30	0.45 ± 0.29	0.44 ± 0.33
		0.8	0.54 ± 0.27	0.52 ± 0.30	0.48 ± 0.26	0.52 ± 0.31	0.43 ± 0.30
24 vs. 24	2 vs. 2	0.6	0.52 ± 0.28	0.49 ± 0.30	0.50 ± 0.30	0.48 ± 0.28	0.55 ± 0.26
		0.8	0.53 ± 0.27	0.56 ± 0.30	0.51 ± 0.30	0.56 ± 0.32	0.52 ± 0.30
	3 vs. 3	0.6	0.48 ± 0.29	0.54 ± 0.27	0.49 ± 0.26	0.49 ± 0.30	0.52 ± 0.30
		0.8	0.55 ± 0.29	0.49 ± 0.27	0.52 ± 0.28	0.49 ± 0.30	0.47 ± 0.26
	6 vs. 6	0.6	0.47 ± 0.30	0.45 ± 0.31	0.51 ± 0.29	0.56 ± 0.28	0.49 ± 0.29
		0.8	0.51 ± 0.30	0.47 ± 0.28	0.54 ± 0.28	0.56 ± 0.31	0.51 ± 0.31

4 Discussion

Discuss the limitation of your method. Discuss what you could have done better for future studies. Discuss what you did not do that you could have done. Often putting incomplete results into discussion is a good idea. You can also provide an alternate strategy in case the proposed method did not perform well.

5 Conclusion

Concluding remarks. State what conclusion you obtained from the research.

Academic Integrity

This course follows the University of Wisconsin–Madison policy on academic integrity. Students are responsible for reviewing and complying with Chapter UWS-

14 at https://docs.legis.wisconsin.gov/code/admin_code/uws/14.

6 References

Provide minimum 10-20 relevant references to your method. The journal paper is cited as Chung et al. (2001) for multiple authors or Songdechakraiwut and Chung (2023) for two author papers. Conference paper is cited similarly as Chung et al. (2003). Books are cited as Chung (2012). See reference below to see how these citations are listed. You should reference more than 10 papers for research paper.

References

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